



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Some important requirements are behind every system.



Learning Outcome:

Make the coding of the system modules and the main program, starting from using the chosen programming language, based on design specifications.

Exchanges information about requirements types and their characteristics.

Can Do Statements:

- I can explain what a functional requirement is.
- I can describe a non-functional requirement.
- I can establish differences between functional and non-functional requirements.

Ask your partner:

- How can you ensure that a system works perfectly? Give some ideas.
- How important are privacy requirements in any type of system? Why?

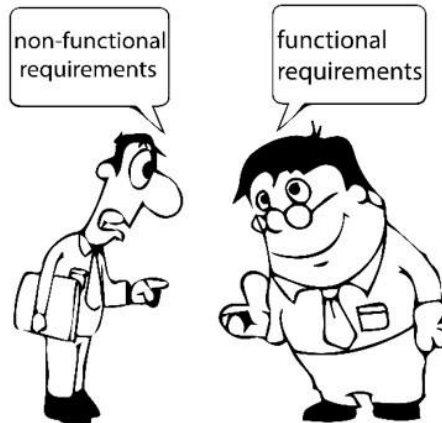




Let's Explore!

This week's task* is about functional and non-functional requirements.

*A "task" is an assignment that you have to do at a given time.



Instructions: Read the text below.

Sarah: Here, this is this week's task. You are not going to have to do much, just check the functional and non-functional requirements of this new system before we launch it.

Peter: Wait! What are those?

Sarah: You've got to be kidding. You have worked on that for weeks! The functional requirements are those that define what the system has to do in specific situations. In other words, that's the one that states the way the input, output, and real performance of a system work together and produce a desired outcome. You're not going to have to focus on other aspects that are not the function of the system and its compliance with the specific objective of the product.

Peter: Got it! What about the other requirement?

Sarah: Once you've assessed the non-functional requirements, you have to make sure they have evidenced the overall operation of the system; in other words, the way the system has to be. You haven't got to forget that you knew them as quality attributes too, as they made reference to aspects like security, privacy, and legislative requirements.

Peter: I haven't got to forget that!

Sarah: Exactly, you definitely haven't got to confuse those two requirements. Let me give some examples. Within the functional requirements, you are not going to have to find other aspects that are not the descriptions of the operations, which the system performs in given cases as well as the reports and the workflow of the system. On the other hand, the non-functional requirements hasn't got to focus on what happens in specific situations but on the legislative, privacy, safety, and ethical requirements of the system. You're not going to have to learn their definitions but to identify the cases you encounter.

**Let's Practice!**

Sara is giving her concept about functional requirements.

Instructions: Use the words from the box to correctly complete the sentences.

worked together / had to use produced / have
worked / have focused

1. You _____ on that for weeks!
2. You _____ functional requirements to define what the system had to do in that specific situation.
3. In other words, that was the one you used to state the way the input and output _____.
4. The real performance of the system _____ the desired outcome.
5. You _____ on other aspects that are not the function of the system and its compliance with the specific objective of the product.

**Let's Practice!**

Peter is a tech-savvy*.

*A *tech-savvy* is a person who knows a lot about current technology.

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: You are not going to have to, You haven't got, are not going to have to, haven't got to forget. Practice saying the sentences aloud. Check your pronunciation with your instructor.

1. _____ to forget that people also know them as quality attributes too.
2. _____ make reference to aspects like security, privacy, and legislative requirements next week.
3. You _____ do much, just check the functional and non-functional requirements of this new system before we launch it.
4. You _____ the way the system had to be.





Extension Activity

Let's have a look at business requirements.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for examples about business functional and non-functional requirements.
- Make a list with those ones that are new for you.
- Make a poster in which you list the requirements.
- Explain to your partners why each requirement is functional or non-functional.

Self-Assessment

Yes

No

- I look for new functional and non-functional examples. ☐ ☐
- I make a poster showing those requirements and its features. ☐ ☐
- I talk about functional and non-functional requirements in business contexts. ☐ ☐



Target Words

- **launch** (verb): to formally present a product to the public
e.g. Just check the functional and non-functional requirements of this new system before we launch it.
- **compliance** (noun): the level in which something is achieved
e.g. You're not going to have to focus on other aspects that are not the function of the system and its compliance with the specific objective of the product.
- **workflow** (noun): the organization in which workers or machines work together to perform a task
e.g. you are not going to have to find other aspects that are not the descriptions of the operations, which the system performs in given cases as well as the reports and the workflow of the system.
- **assess** (verb): to evaluate critically and analytically a situation or object
e.g. Once you've assessed the non-functional requirements, you have to make sure they have evidenced the overall operation of the system.
- **state** (verb): to give clear evidence about something
e.g. In other words, that's the one that states the way the input, output, and real performance of a system work together and produce a desired outcome.
- **outcome** (noun): the final result of a process or situation
e.g. In other words, that's the one that states the way the input, output, and real performance of a system work together and produce a desired outcome.



Learning Points

Have Got to and Going to Have to for What Is Necessary (Negative Form)

Have got to and **going to have** in their negative forms, both represent the idea that something is not necessary, but in different moments in time.

HAVE GOT TO

When you use **"have got to"** in the negative form, the idea that something is not necessary to do arises. You use it for mentioning that there is no need to proceed in a specific way regarding a situation.

The structure that you need to follow to create negative sentences is:

Subject + have/has + not + got to + verb (base form) + complement.

For example,

*These systems **haven't got to be** ready tomorrow, but today!*

*He **hasn't got to include** calculations in the system.*

GOING TO HAVE TO

"Going to have to" represents the same idea as the form **"have got to"**; however, it implies that the action to be taken is required to happen at some point or moment in the future.

The structure that you need to follow is as follows:

Subject + am/is/are + not + going to have to + verb (base form) + complement.

For example,

*You **are not going to have to** fix that bug now.*

*She **is not going to have to** check the data manipulation in the system.*

Examples

- *You **haven't got to** forget that they have been known as quality attributes too as they make reference to aspects like security, privacy and legislative requirements.*
- *You **are not going to have to** do much, just check the functional and non-functional requirements of this new system before we launch it.*





Learning Points

Functional and Non-Functional Requirements

Within the systems engineering, functional and non-functional requirements work together in order to make systems or products operate.

The objectives that a product or system aims to achieve are specifically represented, among others, by the **functional and non-functional requirements**.

FUNCTIONAL REQUIREMENTS

Within the **functional requirements**, information related to the functions and activities that the system or product has to perform in **specific situations** is detailed. In other words, elements such as the input, output, and actual performance of a system concern this field. Furthermore, the functional requirements are not written using technical language as everybody must understand them.

What you find in the functional requirement information is basically descriptions regarding the operations that the system or product performs, such as the work-flow of the system, reports, and regulatory requirements.

NON-FUNCTIONAL REQUIREMENTS

Contrary to the functional requirements, the **non-functional requirements** depict the operation of a system which is ruled by the system architecture. The functional requirement illustrates what a system has to do and react in specific situations, whereas the non-functional requirements illustrate how this system has to be.

The non-functional requirements, or quality attributes, are reflected on elements like the timing, the application domain of the system, the legislative, privacy, safety, and ethical requirements as well as the interoperability.

